

Epigenetic Language Models: Lifelong Learning by Intercalating Knowledge into Structural Zeros

Free-Slot Adaptation on a real 8B ternary MoE

Parallax / Chutes.ai

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AI-assisted draft. Revised through repeated adversarial review; experiments run and checked by the authors. Results are from one real 8B ternary MoE on one 40-fact task, single seed; seeds pending. The ternary Bop repeat uses the same seed, facts and protocol; it does not establish robustness across seeds. The multi-session epigenesis experiment (EP-1) is pre-registered, in progress and untested; the personal-model use case (§8) is an *untested proposal*. Code: github.com/chutesai/epigenesis.

Abstract

An epigenetic language model keeps its base weights as a frozen genome and writes acquired knowledge into a separate sparse layer occupying exact zeros. Removing that layer restores the base bit-for-bit. We test Free-Slot Adaptation (FSA) on a real 8B ternary mixture-of-experts model. On one 40-fact task, ternary Bop with a local output penalty, a 300k-slot budget and rehearsal recalls 79/80 held-out items under both teacher forcing and free-running greedy generation with a clean stop. It fills 249,755 slots, adding no parameters and preserving the packed format, at -0.0001 ± 0.0017 nats/token general-text change. Position-inclusive storage is 0.33 MB entropy-coded or 0.84 MB indexed; the closest LoRA baseline, rank 1 with measured int4 factors, costs 0.69 MB and recalls 75/80 on both metrics. Across tested LoRA ranks and precisions, the recall gap is 1–4 items on one seed: suggestive, not established; seeds pending. No damage difference is detectable between the best penalized FSA runs and fp32 penalized LoRA at this sample size. Rehearsal and damage evaluation use disjoint sequences from the same corpus, so preservation is measured in-distribution. Every run revokes bit-exactly on the checked window. Synthetic packed patches show no detectable serving slowdown; learned patches are not yet benchmarked. FSA does not beat LoRA overall on this evidence. Multi-session concept learning remains untested.

1 Introduction

We use an epigenetic analogy: the base weights are a frozen genome, while acquired knowledge lives in a separate sparse layer that can be removed to recover the base bit-for-bit. Extreme quantization already leaves many weights at exact zero. Can those existing slots carry new knowledge without adding parameters?

Free-Slot Adaptation (FSA). In the pair8 ternary format, at most two of every four aligned pairs are active. Let $\mathcal{S}_0$ be the base’s nonzero support and $\mathcal{Z}_0$ its zeros. Freeze the base and all shared parameters, then train a disjoint slice $\mathcal{R}_d \subseteq \mathcal{Z}_0$ inside already-active pairs. Store its positions and signs as a removable patch. Enabling the patch uses $\mathcal{S}_0 \cup \mathcal{R}_d$; disabling it uses $\mathcal{S}_0$ alone. This is Mallya and Lazebnik [2018]’s mechanism, with the reserve supplied by the compression format rather than a pruning pass.

What we claim. The best observed patch recalls 79/80 teacher-forced and generated items from 249,755 slots (0.33–0.84 MB including positions), with damage within noise of zero. The nearest-size LoRA, rank 1 int4 (0.69 MB), recalls 75/80 on both metrics. A 1–4-item gap across LoRA ranks and measured precisions on one seed is suggestive, not established; seeds pending. No damage difference is detectable between the best penalized FSA runs and fp32 penalized LoRA. Synthetic packed patches have no detectable serving cost, but learned patches still need benchmarking. FSA does not beat LoRA overall; restoration when the patch is removed does not preserve behavior while it is enabled. Multi-session concept learning and the personal-model application remain untested.

Contributions. (1) Quantization zeros as a removable adaptation reserve, with accounting for positions, values, scales, and inference format. (2) Real-model results comparing ternary Bop, learn-then-prune, fp4, and LoRA on recall, damage, and storage. (3) A measured KL floor under random perturbations, with routing near-ties as a possible explanation rather than an established cause. (4) A patch-revocation contract and a pre-registered multi-session concept experiment (§8).

2 Background and related work

Mask-based continual learning. PackNet [Mallya and Lazebnik, 2018] freezes kept weights and trains freed ones, recovering a task by its mask; Piggyback [Mallya et al., 2018] *learns masks over fixed weights* (it does not refill zeros); HAT [Serra et al., 2018] uses hard task-attention and reports *reduced*, not zero, forgetting; Supermasks [Zhou et al., 2019] make the mask itself the trainable object; Progressive Networks [Rusu et al., 2016] add and could drop columns. FSA is PackNet with the free support supplied by the quantization format. **Quantized continual learning.** Ada-QPackNet [Pietroń et al., 2023] and TinySubNets [Pietroń et al., 2025] already combine PackNet-style capacity reuse with quantization; SPDF [Thangarasa et al., 2023] reopens zeroed pretrained weights for dense fine-tuning. These sharply narrow our novelty; our distinct element is a reserve *pre-addressed by the deployed format* and composed by union. **Parameter-efficient and continual adapters.** LoRA [Hu et al., 2022], QLoRA [Detrmers et al., 2023], QA-LoRA [Xu et al., 2024]; Biderman et al. [2024] find LoRA learns and forgets less than full fine-tuning (a possible rank explanation, not a theorem); O-LoRA [Wang et al., 2023] does continual learning in orthogonal LoRA subspaces. **Sparse / composable adaptation.** Diff pruning [Guo et al., 2021] and composable sparse fine-tuning [Ansell et al., 2022]; union of *disjoint* sparse supports is the easy case of the latter. **Editing, memory, unlearning.** MEMIT [Meng et al., 2023] edits facts in place and can override; FSA adds paths that can suppress or correct outputs while enabled but cannot edit a base weight; GRACE [Hartvigsen et al., 2023] and WISE [Wang et al., 2024] keep deletable, routed side-memory for sequential edits while freezing the base; SISA [Bourtole et al., 2021] is exact unlearning by shard isolation; TOFU [Maini et al., 2024] benchmarks fictitious-fact forgetting. **Personalization and retrieval.** OPPU [Tan et al., 2024] attaches a per-user PEFT module (removed by user identity, not per record); LaMP [Salemi et al., 2024] is a personalization benchmark with retrieval baselines; RAG [Lewis et al., 2020] and kNN-LM [Khandelwal et al., 2020] store facts externally and delete them by dropping a record—the baseline any “delete a fact” claim must beat. **Low-bit models.** BitNet b1.58 [Ma et al., 2024, 2025] and LittleBit [Lee et al., 2025]. Bop [Helwegen et al., 2019] optimizes binary weights without latent real weights; we extend its gradient-memory rule to ternary birth and death (§7).

3 Method

3.1 Format and reserve

In pair8, weights form aligned blocks of eight (four pairs); each weight is ternary times a scale, and at most two pairs per block are non-zero. A block thus has 417 reachable states ($1 + 4 \cdot 8 + \binom{4}{2} 8^2$), i.e. $\log_2 417/8 \approx 1.088$ bits/weight asymptotically (1.125 at a fixed 9 bits/block), and is $\geq 50\%$ zero *by construction* (ternary alone guarantees no such fraction). **Writable reserve.** Zeros inside already-active pairs can be filled while preserving pair occupancy and the codebook. At two active pairs, opening a third pair is illegal, but filling a zero member of an active pair is legal. Thus the writable in-codebook reserve is a subset of the full zero support, not necessarily empty in a full-occupancy block. With pair occupancy frozen, opening new pairs requires a separate sidecar patch outside the base codebook. Filled positions and signs must still be stored for revocation; the joint rate of 1.088 bits/weight is not the marginal price of a filled slot.

3.2 Adaptation and exact revoke

Train only the free-slot patch ΔW , with support $\mathcal{R}_d \cap \mathcal{S}_0 = \emptyset$, while the base remains live. Base weights, per-row scales, pair occupancy, router, and norms are frozen. A legal ternary writer fills free slots with $\pm\alpha$, where α is the frozen row scale. The enabled model uses $W + \Delta W$; zeroing the filled slots returns every patched tensor bit-identical to W , since the patch never overwrites base support or shared state. In every run, patch-off logits were bitwise equal to the base on one 512-token held-out window, and test recall returned to 0/80.

This restores the base, without guaranteeing its behavior while the patch is on. It does not delete individual facts within a patch. If later sessions train with an earlier patch enabled, zeroing that session’s slots alone need not remove its influence on retained patches or training state (§8).

3.3 Honest accounting

Plain ternary FSA adds **zero new logical parameters**: the element count is unchanged. It fills only zeros inside active pairs, so a merged patch uses the same packed tensors and adds no tensors or inference compute. This claim depends on preserving pair occupancy and the serving format; opening new pairs or using a separate patch can add work. Synthetic legal patches have been benchmarked in packed serving (§6.4); learned patches have not. Storage is not free: revocation needs positions and signs in a stored diff, or the original pack. The learned-block-scale arm adds parameters and storage and is outside these packed-format claims. A full per-domain mask would cost one bit

per weight; sparse position coding avoids that cost for sparse patches. Frozen row scales constrain each ternary slot to $\{-\alpha, 0, +\alpha\}$, a capacity restriction. Actual position-inclusive patch sizes are reported in §6; hypothetical trit-only accounting is insufficient.

4 Real-model setup

Lambda is an 8B-parameter ternary MoE with 128 routed experts per MoE layer, 12 active per token, model width 1152, and a recurrent sequence-mixing state between layers. Each routed expert has a 2304×384 up projection, squared ReLU, and a 384×2304 down projection, stored as `pair8` trits times frozen per-row scales. We patch layers 14–15, all 128 experts, both projections: a broad footprint with 66,807,179 free slots inside already-active pairs. Base support, scales, pair occupancy, router, and all other parameters remain frozen. Plain ternary births take value $\pm\alpha$ for row scale α . The block-scale arm keeps ternary signs but multiplies them by learned magnitudes, one per row and 16 input columns, initialized at 0.5α . Its values leave $\{-\alpha, 0, +\alpha\}$; the packed-format and no-new-parameter claims do not cover this arm. The fp4 variant uses magnitudes $\{0, 0.125, 0.25, 0.375, 0.5, 0.75, 1, 1.5\}\alpha$ (the E2M1 grid scaled by 0.25α), with minimum nonzero magnitude 0.125α and cap 1.5α ; it leaves the ternary codebook. Evaluation decodes experts to bf16. Legal ternary patches could use the same packed kernel; synthetic patches have been benchmarked there (§6.4), while accuracy evaluation still used decoded bf16 weights.

The task has 40 synthetic facts about a fictional person (favorite cheese, boat’s name, house number). Training lasts 300 steps with answer-token-only loss. Each step samples three phrasings per fact from a pool of 12 templates (three plain sentences and nine augmented forms, including “Question:/Answer:” and “User:/Assistant:”). Two further phrasings and 16×512 general-text windows provide validation monitoring and checkpoint selection. Test recall is scored under teacher forcing: given the prompt, the model’s greedy (argmax) prediction must match every answer token. The new runs also measure free-running greedy generated recall: the whole answer must match at a word boundary, then a terminator or end-of-sequence within two tokens (text after that is ignored). Older runs test teacher-forced recall only. Recall is measured on two phrasings never used in training or validation: “Q: What is X’s attr? A:” and “A little-known fact about X: her attr is”. The Q/A form is held out verbatim but resembles trained question/answer formats. There are 80 test items; one item is 1.25 percentage points, and the unpatched model gets 0/80. Damage is mean per-token Δ NLL relative to the base on 40 disjoint 512-token WikiText windows (about 20k tokens), with paired standard error (SE) over windows. The unpatched model’s NLL is 2.890 on a separate small diagnostic pool of five 512-token held-out windows, which also provides the reference for the chaos-floor NLL ranges. All results use one seed per run; same-configuration repeats do not test robustness across seeds. Seeds pending for the headline configuration. Table 2 reports step 300, except repair, which adds 50 steps with births disabled.

5 A chaos floor for KL

Random, untrained ternary flips produce a KL floor while NLL on the separate five-window diagnostic pool stays within about 0.01 nats of the unpatched model. For the down-projection flips in Table 1 and Figure 1, the range is about -0.010 to $+0.003$ relative to the diagnostic-pool base NLL of 2.890; across all flip settings tested, it is about -0.010 to $+0.005$. A single layer-15 zero in one expert’s down projection illustrates the bf16 threshold: changes at most 0.01α round away; a full trit changes about 6% of top-1 tokens. Two different single flips disagree with each other by KL 0.027, slightly more than one flip differs from the base (0.024), and the divergence is roughly flat across sequence position. These observations suggest a different but equally good trajectory after a change exceeds bf16 resolution. Near-ties in discrete MoE routing, propagated through recurrent state, are a plausible driver, not a proven explanation.

For always-on edits the typical KL floor is about 0.02–0.027, with layer-dependent exceptions in Table 1. A KL penalty mostly fights this floor: increasing its weight from 1 to 30 left KL near 0.06 while recall fell from 48% to 20%. We therefore measure damage with paired Δ NLL, interpret KL relative to this floor, and use a local output penalty for preservation (§7).

6 Results

6.1 Recall and damage

The earlier unbudgeted Bop + penalty run recalled 80/80 under teacher forcing, but its same-configuration rerun recalled 78/80 TF and 78/80 GEN at -0.0011 ± 0.0019 damage. The 80/80 result was not reproduced exactly: variation between executions is a couple of items on one seed. The two plain Bop $\tau = 0.7$ executions also use the same seed and protocol; the repeat listed decay, but Bop has no real-valued weights to decay, so it is the same

Random flips	Layer 2 KL	Layer 15 KL	Layer 28 KL
1	0.027	0.023	0.007
1,000	0.027	0.023	0.015
1,000,000	0.028	0.026	0.023

Single change	KL	Top-1 agreement
$\varepsilon \leq 0.01$	0	1.000
$\varepsilon = 0.1$	0.0105	0.978
$\varepsilon = 1$	0.0238	0.939

Table 1: Untrained perturbations (single seed). Top: random free-slot flips in down projections. Bottom: one layer-15 weight changed by $\varepsilon\alpha$; small changes are absorbed by bf16 rounding.

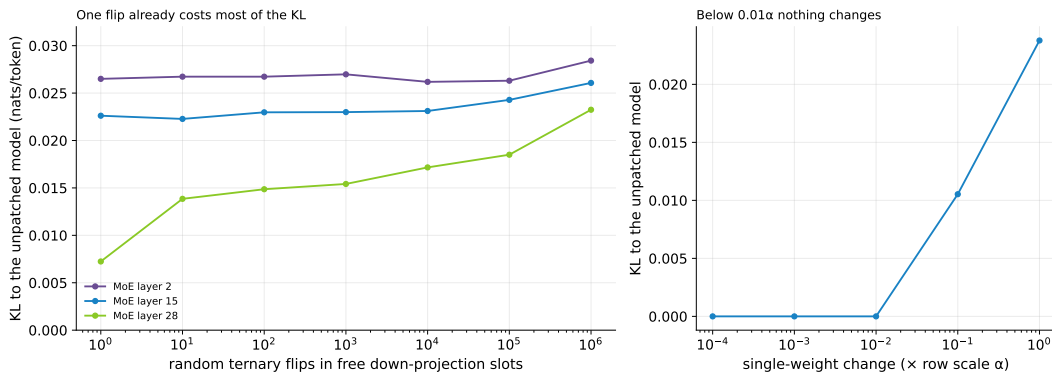


Figure 1: KL against the number of random ternary flips in free down-projection slots at layers 2, 15, and 28 (left), and against a single-weight change relative to its row scale (right). Unlike layers 2 and 15, layer 28 has a lower KL of 0.007 for a single flip, rising to 0.023 at one million flips. NLL on the separate diagnostic pool of five 512-token held-out windows remains within about 0.01 nats of its base NLL of 2.890 (range about -0.010 to $+0.003$) for these down-projection flips; across all flip settings tested, the range is about -0.010 to $+0.005$.

configuration. This variation does not establish robustness across seeds. Earlier validation-selected checkpoints remain distinct from the final endpoints reported here.

6.2 Storage

6.3 Honest reading

At a comparable byte budget, the best FSA patch recalls 79/80 TF and 79/80 GEN from 0.33 MB entropy-coded or 0.84 MB indexed. Across every LoRA rank and measured precision tried, ranks 1–4 recall 75–78/80 TF and 75–77/80 GEN at 0.69–11.01 MB (int4 to bf16). The nearest LoRA size is rank 1 int4, 0.69 MB: 75/80 on both metrics versus 79/80 for FSA. The two FSA encodings bracket that size; this is a comparable budget, not exact byte equality. The gap is 1–4 items on one seed: suggestive, not established; seeds pending. FSA does not beat LoRA overall on this evidence.

The best penalized FSA runs, with or without rehearsal, and fp32 penalized LoRA all have $|\Delta\text{NLL}| \leq 0.003$ with SE around 0.002: at this sample size no damage difference is detectable between them. This summary does not cover rank-1 bf16/int8 evaluations, whose measured damage is $+0.0040/+0.0049$ (Table 3). Plain Bop without the penalty costs $+0.024$ to $+0.031$; those rows remain as history. Rehearsal uses the same corpus as the damage panel, with no shared 32-token sequences, so near-zero damage is measured in-distribution; other kinds of text are untested. Removing the patch restores the base exactly, without establishing preservation while enabled. Multi-session concepts remain untested.

Method	TF (of 80)	GEN (of 80)	$\Delta\text{NLL} \pm \text{SE}$
Bop + penalty + budget + rehearsal (best)	79	79	-0.0001 ± 0.0017
Bop + penalty + budget + energy threshold	79	78	$+0.0013 \pm 0.0019$
Bop + penalty + budget + repair	78	78	$+0.0029 \pm 0.0016$
Bop + penalty + budget	77	76	-0.0003 ± 0.0021
Bop + penalty (rerun)	78	78	-0.0011 ± 0.0019
Bop + penalty (earlier)	80	—	$+0.0015 \pm 0.0017$
Bop $\tau = 0.7$ (run 1)	80	—	$+0.0266 \pm 0.0035$
Bop $\tau = 0.7$ (repeat)	80	—	$+0.0237 \pm 0.0036$
Bop $\tau = 0.9$	80	—	$+0.0314 \pm 0.0033$
Learn-then-prune, decay 0.3	73	—	$+0.0250 \pm 0.0032$
Learn-then-prune, decay 0.5	71	—	$+0.0129 \pm 0.0028$
Learn-then-prune, decay 0.1	75	—	$+0.0667 \pm 0.0035$
Signs + learned block scales, decay 0.3	70	—	$+0.0282 \pm 0.0031$
Fp4 + penalty	77	—	$+0.0050 \pm 0.0015$
LoRA $r = 1$ + penalty, fp32 (baseline)	75	76	-0.0002 ± 0.0019
LoRA $r = 2$ + penalty, fp32 (baseline)	77	76	$+0.0019 \pm 0.0014$
LoRA $r = 4$ + penalty, fp32 (baseline)	77	76	-0.0001 ± 0.0020
LoRA $r = 16$ + penalty (baseline)	78	—	$+0.0005 \pm 0.0018$
LoRA $r = 16$, unregularized (baseline)	78	—	$+0.9361 \pm 0.0302$

Table 2: Single-seed final endpoints on the real model. TF: teacher-forced recall; GEN: generated recall with a clean stop; —: not measured. All FSA rows use ternary slots except fp4. The budget is 300k slots; FSA penalty is $\lambda = 10$, LoRA penalty $\lambda = 100$. Repair reports 300 training plus 50 death-only steps; all other endpoints are step 300. Every run restores bitwise-equal logits on one 512-token window and test recall to 0/80. Seeds pending.

LoRA baseline	Precision	TF (of 80)	GEN (of 80)	$\Delta\text{NLL} \pm \text{SE}$
$r = 1$ + penalty	bf16	76	75	$+0.0040 \pm 0.0020$
$r = 1$ + penalty	int8	76	76	$+0.0049 \pm 0.0018$
$r = 1$ + penalty	int4	75	75	$+0.0021 \pm 0.0017$
$r = 2$ + penalty	bf16	76	77	$+0.0018 \pm 0.0017$
$r = 2$ + penalty	int8	77	76	$+0.0013 \pm 0.0014$
$r = 2$ + penalty	int4	78	76	$+0.0021 \pm 0.0017$
$r = 4$ + penalty	bf16	78	76	-0.0010 ± 0.0018
$r = 4$ + penalty	int8	77	76	$+0.0015 \pm 0.0017$
$r = 4$ + penalty	int4	78	77	$+0.0003 \pm 0.0016$

Table 3: Measured factor-precision evaluations of the same single-seed LoRA adapters, without retraining. Int8/int4 are fake-quantized, with one bf16 absmax scale per rank vector. Quantized accuracy and damage were measured; seeds pending.

6.4 Serving

Synthetic legal patches use uniformly drawn slots with random signs at the size of the earlier Bop patches (670k) and a 3M stress test. They were written into the released packed model and benchmarked on Mac Metal, Mac CPU and a Snapdragon 8 Elite phone; learned patches (now about 250k slots) have not been benchmarked. The table gives paired patched/base throughput medians. Paired median ratios stay within about 2% of 1.0, with no consistent direction; ranges are min–max over rounds. For Metal text-prefill, the ranges touch 1.0 ($[0.993, 1.000]$ and $[0.991, 1.000]$). The Mac model file is 2,133,006,592 B and the phone’s Q8-trunk file is 1,656,608,512 B; patching leaves each file’s size unchanged. Peak memory is unchanged to reported precision, about 2.75 GiB on Metal, 3.29 GiB on Mac CPU and 3.24 GB on the phone. Phone readings span 3.244–3.245 GB. Revocation restores byte-identical files, checked by SHA-256. Accuracy evaluation still used decoded bf16 weights. Synthetic patches test kernel cost, not learned-patch accuracy or routing effects. LoRA is an analytic estimate only in this runtime: rank 16 would

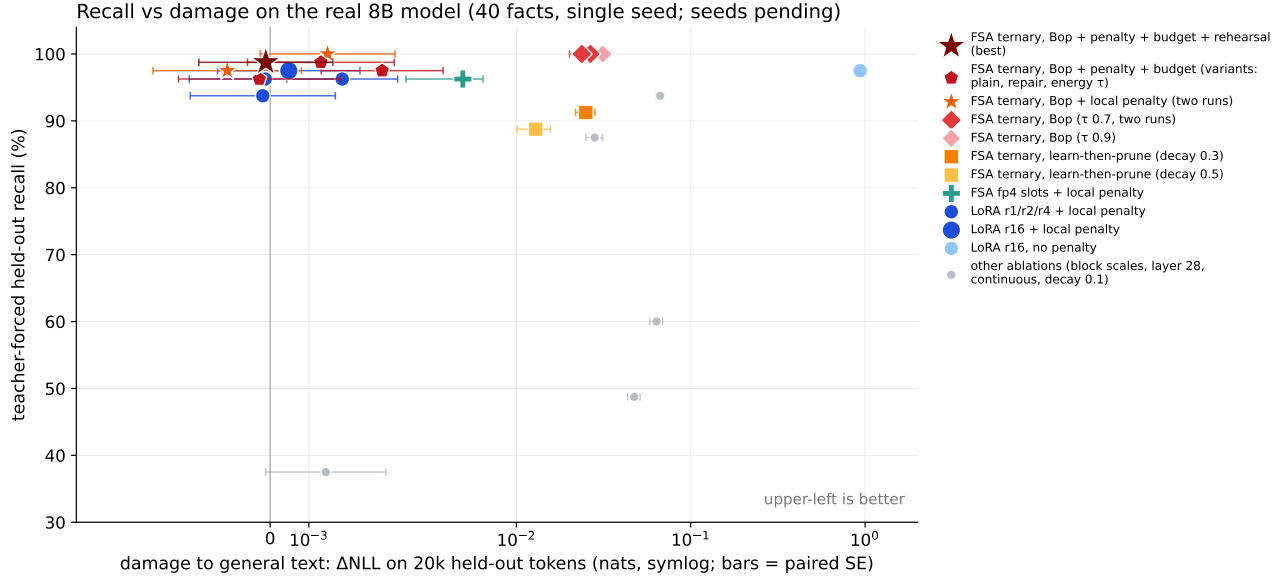


Figure 2: Teacher-forced held-out recall versus general-text Δ NLL, with paired-SE bars. The symmetric log axis is linear near zero, preserving negative estimates and their uncertainty. All points use one seed, 12 training formats, two held-out phrasings and 40 disjoint 512-token WikiText windows. Minor ablations include block scales and layer-28 down-only patches. Seeds pending.

Patch	Slots / parameters	E MB	I MB
Bop + penalty + budget + rehearsal (best)	249,755	0.33	0.84
Bop + penalty + budget + energy threshold	238,662	0.32	0.81
Bop + penalty + budget + repair	250,538	0.33	0.85
Bop + penalty + budget	243,935	0.32	0.82
Bop + penalty (rerun)	735,955	0.82	2.48
Bop + penalty (earlier)	739,549	0.83	2.50
Bop $\tau = 0.7$ (run 1)	812,420	0.89	2.74
Bop $\tau = 0.7$ (repeat)	669,705	0.76	2.26
Bop $\tau = 0.9$	588,673	0.68	1.99
Learn-then-prune, decay 0.3	542,205	0.64	1.83
Learn-then-prune, decay 0.5	523,229	0.62	1.77
Learn-then-prune, decay 0.1	1,857,504	1.76	6.27
Signs + learned block scales, decay 0.3	1,125,769	1.17 + scales	3.80 + scales
Fp4 + penalty	21,265,606	18.17	79.75
LoRA $r = 1$ (baseline)	1,376,256	2.75 / 1.38 / 0.69	
LoRA $r = 2$ (baseline)	2,752,512	5.51 / 2.76 / 1.38	
LoRA $r = 4$ (baseline)	5,505,024	11.01 / 5.51 / 2.76	
LoRA $r = 16$ (baseline)	22,020,096	44.04 bf16	

Table 4: Position-inclusive estimates, decimal MB (single seed per run). E: entropy-coded positions plus signs over 66,807,179 free slots; I: 26-bit index plus value (1 bit ternary, 4 bits fp4). LoRA ranks 1–4 list bf16 / int8 / int4, including bf16 scales for int8/int4, with measured accuracy at each precision. Rank 16 lists bf16 size with accuracy of fp32 factors; no bf16 evaluation is available. Block scales add about 113 MB. Fp32 training factors use twice the bf16 bytes.

add about 0.5–1.5% decode cost and 44 MB per adapter; merging would densify 64 MB of packed experts to 906 MB bf16. Switching a running model’s patch is not yet implemented; today switching requires changing the packed model and reloading. One model copy serves one user’s patch at a time, unlike multi-adapter LoRA batching.

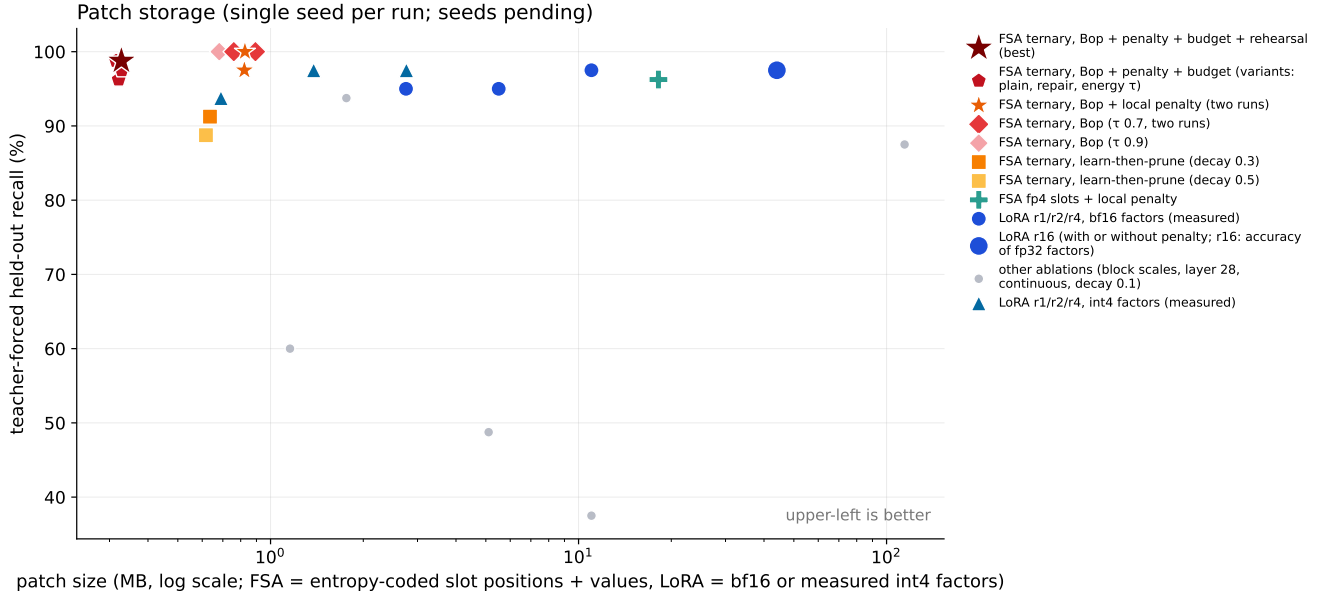


Figure 3: Teacher-forced recall versus position-inclusive storage (single seed per run; seeds pending). FSA uses entropy-coded positions and values, over each run’s own reserve. LoRA ranks 1–4 show measured bf16 recall (95%, 95%, 97.5%) at bf16 size and measured int4 points; r16: accuracy of fp32 factors, shown at bf16 size because no bf16 evaluation is available. Minor layer-28 ablations have a 16.8M-slot reserve; learned block scales add about 113 MB in fp32, bringing that ablation to about 114 MB.

Device / backend	Decode, 670k	Decode, 3M	Prefill, 670k	Prefill, 3M
Mac M5 Max, Metal	0.999	0.999	0.999	1.001
Mac M5 Max, CPU, 8 threads (table lookup)	1.005	0.990	1.002	1.009
Mac M5 Max, CPU, 8 threads (alternate lookup)	1.006	0.998	1.004	0.999
Snapdragon 8 Elite, CPU, Q8 trunk	1.002	0.982	0.996	1.000

Table 5: Synthetic packed-patch serving ratios, paired within rounds. Median ratios stay within about 2% of 1.0, with no consistent direction. No slowdown was detected; min–max ranges over rounds are not a significance test and do not prove zero cost. Learned patches are not yet benchmarked.

7 Writers

7.1 Ternary Bop

Bop is Helwegen et al.’s binary optimizer [Helwegen et al., 2019]. Our ternary extension adds the birth and death rules below and resets the gradient EMA after every change. Each free slot holds $q_j \in \{-1, 0, +1\}$ directly, with no latent real weight. For loss gradient g_j , maintain

$$m_j \leftarrow (1 - \gamma)m_j + \gamma g_j, \quad \gamma = 0.05.$$

If $q_j = 0$ and $|m_j| > \tau$, birth sets $q_j \leftarrow -\text{sign}(m_j)$. If $q_j \neq 0$, $|m_j| > \tau$, and $\text{sign}(m_j) = q_j$, death sets $q_j \leftarrow 0$: the gradient says the current sign hurts. Reset m_j to zero after each change. Thresholds are 0.7 or 0.9 times the RMS of the first answer-loss gradient over free slots. Historical plain Bop runs use no penalty; new runs add the local penalty below. Bop does not use weight decay. The threshold controls flips: small gradients need not become full $\pm\alpha$ steps, unlike Adam’s per-coordinate normalization, which can turn persistent small gradients into full flips. In the historical single-seed runs, slots are filled early: 812,077 slots were filled at step 50 (812,420 at step 300) at $\tau = 0.7$. Higher τ uses fewer slots (589k) at slightly higher damage, so it is not a simple damage dial. The $\tau = 0.7$ repeat has 669,465 filled slots at step 50 and 669,705 at step 300; its revoke is also bit-exact, with test recall returning to 0/80.

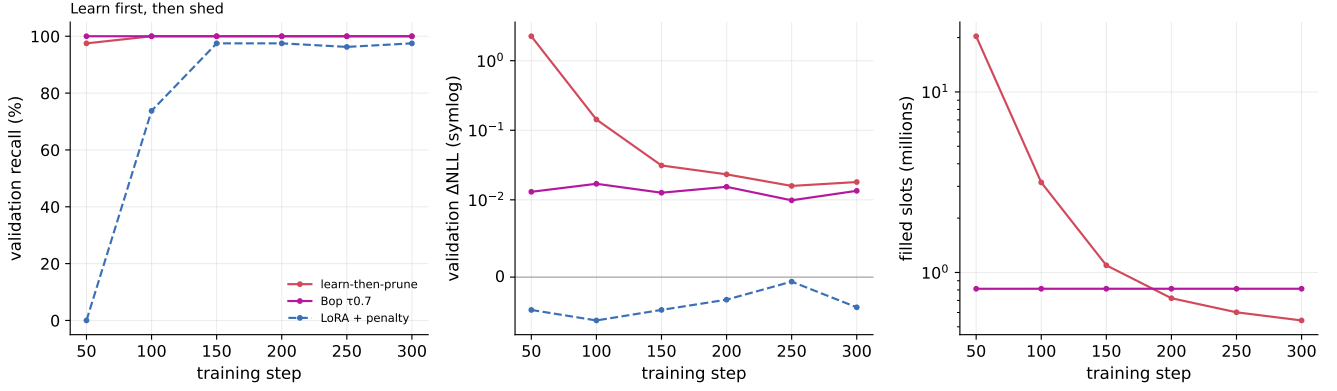


Figure 4: Validation recall, validation ΔNLL (symmetric log scale), and filled slots (millions, log scale) across training. The recall and damage panels show learn-then-prune, ternary Bop $\tau = 0.7$ run 1, and regularized LoRA; learn-then-prune uses decay 0.3. The filled-slot panel shows only those two FSA writers; the new repeat and decay 0.5 trajectories are not plotted. Bop’s filled-slot count is set by its threshold early and then stays flat, while learn-then-prune fills many slots first and sheds them over training. Validation trajectories differ from the held-out test endpoints in Table 2. Each trajectory uses one seed; seeds pending.

7.2 Learn-then-prune

These historical comparisons use a single seed per run. Each free slot has an fp32 master w_j in row-scale units. Forward uses $q_j = \text{sign}(w_j)\mathbf{I}(|w_j| > 0.5)$, where $\mathbf{I}$ is the indicator; gradients pass straight through [Bengio et al., 2013]. AdamW [Loshchilov and Hutter, 2019], learning rate 0.02, applies decoupled weight decay 0.3 to masters. The model learns early: at step 50 validation recall is 97.5%, with 20.3M filled slots and validation $\Delta\text{NLL} +2.2$. Decay pulls weakly supported masters under threshold: 3.2M slots at step 100, 1.1M at 150, and 0.54M at 300. Validation damage falls to about +0.02 and validation recall stays 100%; test recall is separately 91.25%. Held-out damage fell from +0.0282 at step 250 to +0.0250 at step 300, while validation damage rose slightly (+0.016 $\rightarrow$ +0.018). With decay 0.5, filled slots fall from 14.18M at step 50 (98.75% validation recall) to 1.12M at step 100, 0.62M at 150, 0.50M at 200, and 0.45M at 250 (92.5% validation recall), then rise to 0.52M at 300 (97.5% validation recall). Test recall at step 300 is 88.75%; revoke is bit-exact. Whether longer pruning helps is untested.

7.3 Fp4 and the local output penalty

Four-bit float slots times α use AdamW decay 0.1 and add the general-text penalty

$$\lambda \frac{\sum_{l,t} \|x_t \Delta W_l^T\|^2}{\sum_{l,t} \|x_t W_l^T\|^2 + 10^{-12}}, \quad \lambda = 10,$$

where l ranges over patched linear projections, t over general-text tokens, x_t is the input to that projection, W_l is its frozen base weight, and ΔW_l is the patch. The denominator, base-weight output energy on the current inputs, is treated as a constant normalizer and is not differentiated. This smooth local penalty measures each patched projection’s direct output change rather than the change in final logits, so it is not dominated by the whole-model divergence behind the KL floor. Its inputs still come from the patched forward pass: earlier patched layers can change later inputs and routing. The best LoRA run uses the same penalty with $\lambda = 100$. At that strength the penalty stopped earlier ternary learning: its gradient is zero at zero, yet one birth costs a full $\alpha^2 \mathbb{E}[x^2]$. Bop with $\lambda = 10$ now learns while keeping measured damage within noise of zero (Table 2).

7.4 Budget and preservation variants

Gain-minus-cost budget. At most 300k slots may be filled. The allowed set is re-chosen every 10 steps up to step 200 (the harness default; that setting is not stored in the result files), by each slot’s gradient evidence minus its expected general-text cost, α^2 times the input column’s general-text energy. Births go where evidence for the facts is strongest relative to expected general-text change. **Rehearsal.** The headline objective is 0.8 times fact loss plus 0.2 times ordinary next-token loss on general WikiText windows, alongside the local penalty. Those windows share no 32-token sequences with the damage panel but come from the same corpus. Near-zero damage is therefore measured

in-distribution; other kinds of text remain untested. **Energy-scaled threshold.** The row’s birth threshold is multiplied by $\text{clamp}((e_{\text{row}}/\text{median } e)^{0.5}, 0.25, 4)$, where e_{row} is the expected general-text output energy of one birth in that row and the median is over rows with free slots and positive energy. Sensitive rows need more evidence. **Repair.** After 300 training steps, 50 more Bop steps keep the local penalty at $\lambda = 10$ and disable births, allowing deaths only. All variants use one seed; small recall differences do not establish their relative value. Seeds pending.

7.5 Ingredients that mattered

Answer-only loss avoids learning template boilerplate, which caused most early damage with whole-sentence loss. Twelve diverse formats, including Q/A, largely closed the validation-to-test gap (Bop and penalized LoRA test at their validation recall; learn-then-prune decay 0.3 remains 100% validation vs 91.25% test). An earlier protocol reached only 54%/56% held-out recall. The broad footprint mattered: 32 selected experts in one layer, down only, with a 500k-slot budget (350k–404k slots filled) starved capacity (19–29% recall). Damage uses the roughly 20k-token held-out pool, with checkpoint selection on a separate validation pool.

7.6 What did not work

KL penalties fight the chaos floor and cost recall. Ternary slots with smooth penalty and decay peak in recall, then are pruned away. Tiny fp4 slots in late layer 28, down only, with a 2M-slot budget and grid magnitudes starting at 0.03125α , fill about 2.0M slots but yield 0% recall at +0.0067 damage: the patch changed the model but learned none of the tested answers. Patch-only learned block scales are no better than plain ternary (87.5%/+0.028) and add substantial storage. Single-late-layer continuous bf16 slots give 48.75%/+0.047, versus same-placement LoRA’s 37.5%/+0.0014: a recall–damage trade, not an overall improvement over LoRA. Gain-minus-cost ternary slot selection at that placement gives 60%/+0.064.

8 Toward epigenesis: multi-session learning

The ternary reserve now carries facts on a real model (§4). A revocable domain patch is therefore demonstrated for this task: enable the slice for its domain; disable it to recover the base. The personal-model application remains an untested proposal, with a narrower contract:

Patch-revocation contract. Deleting an isolated patch exactly removes *that patch’s parameter contribution*. Equivalence to a model trained without its data additionally requires that no retained patch or training state depended on it. It does not delete individual facts (facts within a slice are superimposed), and it does not make information already in the base inaccessible.

Retrieval deletes individual records while leaving the base unchanged [Lewis et al., 2020, Khandelwal et al., 2020]; disabling a per-user adapter removes that user’s contribution [Tan et al., 2024]. The case for weights is skills, style, and policy that benefit from adaptation in every forward pass. Arbitrary facts remain a retrieval baseline’s natural task. The real ternary fact experiment offers smaller patches, with no detectable damage difference between the best penalized FSA runs and fp32 penalized LoRA at this sample size. General personalization remains untested.

The pre-registered EP-1 experiment is in progress and explicitly **untested**: five sessions with a synthetic user containing 40 policies, 100 recurrent facts (each in 3 of 5 sessions), and 100 one-off “lure” facts (20 per session), consolidating self-study into free ternary slots. An identifiability gate requires the in-context ceiling to beat the unpatched model by at least 10 concept accuracy points on development probes; otherwise nothing is scored. Success criteria fixed in advance require, after session 5, concept gain of at least 10 concept accuracy points over the unpatched model on held-out probes, lure recall no more than 5 accuracy points above it, and mean general-text ΔNLL at most +0.01 nats/token on a locked 40×512 WikiText panel, disjoint from the selection pool, with a paired 95% damage interval excluding +0.05. Final concept accuracy must retain at least 80% of the best earlier session’s accuracy, with no upward ratchet in lure recall (monotone non-decreasing with a net increase).

The pre-registered ternary arm (“ternary with block scales”) uses a 2M-slot ceiling and learned block scales initialized at 0.25α , so like the block-scale arm here it is outside the packed-format / no-new-parameter claims. A separate fp4 arm is also registered; the comparator is regularized LoRA at the same placement.

The pre-registered revoke criterion clears the patch, scales, optimizer and training state, replay, and session ledger. It requires every patched weight tensor to be bitwise equal to the base and format-legal, logits on 32 fixed prompts within 10^{-2} of the base, and concept/lure metrics back to the base within noise. Exact logit equality is reported separately; it was observed when switching patches off in the fact experiment (§4). “Match LoRA” means within 3 concept accuracy points and no more than +0.005 extra damage, with the lure constraint held and revoke passing; “beat LoRA” means at least 5 more concept accuracy points with damage no worse than LoRA. Per the pre-registration, one seed is a pilot; three seeds are needed to claim. Neither multi-session concept learning nor on-device

training is established: optimizer state can be large even when a final patch is small. A broader evaluation should include local retrieval and per-user adapters on personalization benchmarks [Salemi et al., 2024], with unlearning measured [Bourtole et al., 2021, Maini et al., 2024].

9 Limitations

Damage and recall. The best penalized FSA runs and fp32 penalized LoRA are within noise of zero on 40 WikiText windows, without a detectable damage difference at this sample size. Plain Bop costs +0.024 to +0.031. Damage on other corpora and broader behavior is untested; rehearsal windows are sequence-disjoint but drawn from the same corpus as the damage panel. Generated recall is now measured for the new runs, checking unaided answer production and a clean stop; older runs remain teacher-forced only.

Evaluation scope. We have one task, 40 facts about one fictional person, 80 test items, and one seed per run. Same-configuration repeats vary by a couple of items and are not evidence of robustness across seeds. A 1–4-item recall gap is suggestive, not established; seeds pending for the headline configuration. The Q/A test form resembles training; writer and regularization choices differ. The chaos-floor explanation is a hypothesis. The roughly 20k-token damage pool cannot establish broad behavioral safety. Multi-session concept learning remains untested in EP-1.

Revoke scope. Removing the whole patch restores the base bit-for-bit. This does not guarantee preservation while the patch is on, deletion of individual facts, removal of information already in the base, or removal of an earlier session’s influence after later sessions trained on top of it. The frozen base can be augmented or suppressed through new paths but its existing weights cannot be refined.

Format and deployment. Plain ternary writers stay legal pair8, but evaluation decoded experts to bf16; packed-kernel serving is measured for synthetic legal patches only, with learned patches not yet benchmarked. Fp4 and learned-block-scale variants leave the original codebook. Optimizer state can be large even when the final patch is small, and on-device training is not established.

Privacy. Knowledge in weights can be elicited. A per-user model is the proposed deployment for the personal application; its safety is untested, and the patch mechanism supplies no privacy guarantee.

10 Conclusion

A frozen weight genome can acquire a removable sparse layer in its exact zeros. On a real 8B ternary MoE, Bop with a local penalty, a slot budget and rehearsal recalls 79/80 teacher-forced and generated items from 249,755 slots (0.33–0.84 MB including positions) at -0.0001 ± 0.0017 damage. The nearest-size LoRA, rank 1 int4 at 0.69 MB, recalls 75/80 on both metrics. Across tested ranks and precisions, a 1–4-item gap on one seed is suggestive, not established; seeds pending. No damage difference is detectable between the best penalized FSA runs and fp32 penalized LoRA at this sample size, on the same corpus used for rehearsal. FSA does not beat LoRA overall on this evidence. Synthetic packed patches have no detectable serving cost and revoke byte-identically; learned-patch serving, broader preservation and multi-session concept learning remain untested. The pre-registered EP-1 experiment is in progress.

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